

Urban flooding and population exposure analysis in Jakarta

A GIS-Based Assessment of Flood Risk, Drainage Infrastructure Gaps, and Vulnerable Districts

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Course: Spatial Data Science

1. Introduction

- Jakarta faces recurring urban flooding due to dense population, river systems, and low-lying terrain.
- This project uses GIS and spatial analysis techniques to map flood risk across Jakarta districts.
- Multiple spatial factors including flood history, elevation, population, and drainage infrastructure were analyzed.
- The study identifies high-risk and infrastructure-deficient areas

2. Problem Statement

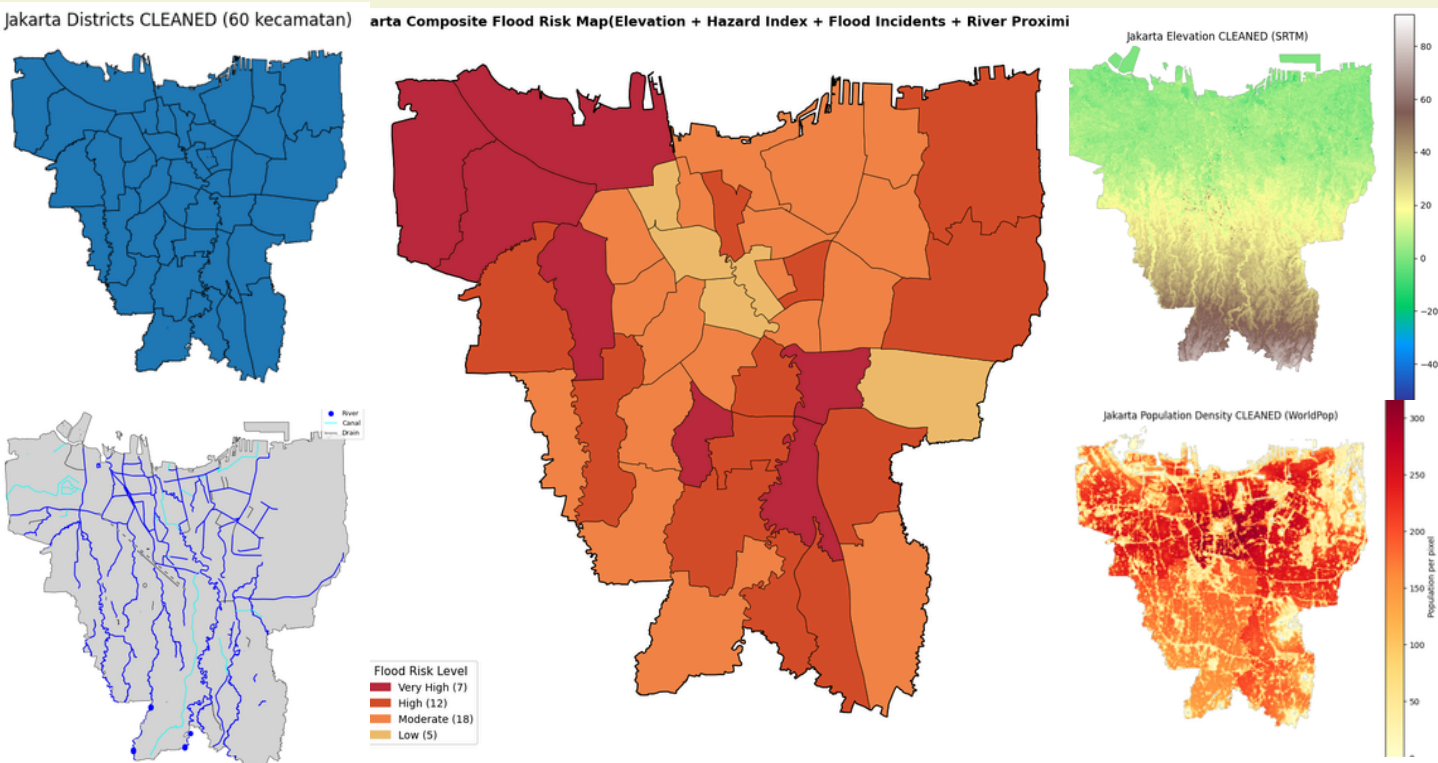
- Which districts in Jakarta are most vulnerable to urban flooding?
- How are flood risk patterns related to population exposure and drainage infrastructure coverage?
- Which high-risk districts should be prioritized for future drainage infrastructure and shelter improvements?

3. Data

- SRTM Elevation Raster** - NASA EarthExplorer
- Population Density Raster** - WorldPop
- Flood Hazard Index 2023** - DKI Jakarta Open Data Portal
- Flood Incident Records 2016–2020** - DKI Jakarta Open Data Portal
- District Boundaries** - OpenStreetMap via OSMnx
- Drainage/River Network** - OpenStreetMap via Overpass API

3.1 Cleaning

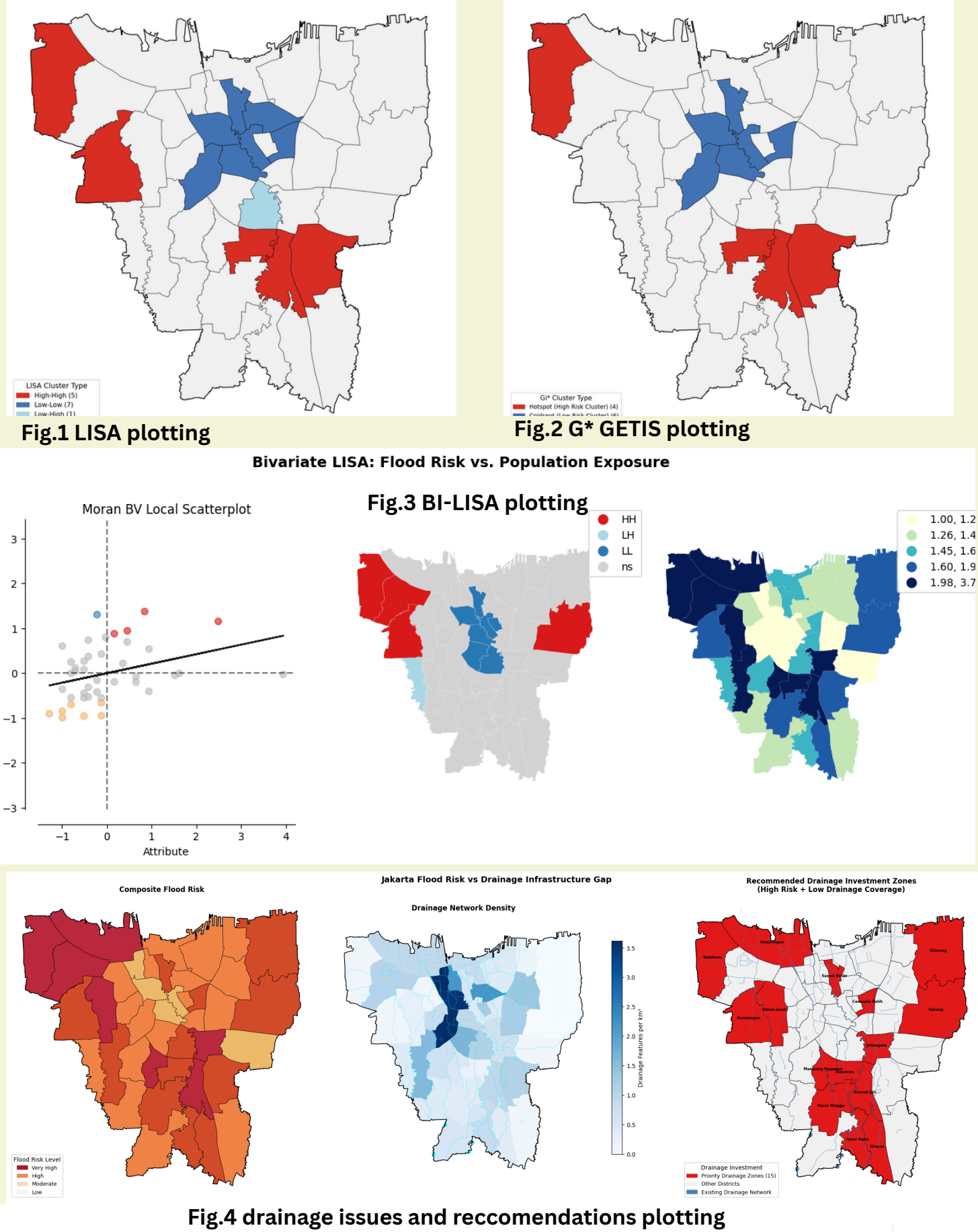
- All datasets were reprojected to the common projected coordinate reference system **EPSG:3857** for meter-based distance calculations.
- Invalid** and **NoData** raster values were masked and removed using null assignment **np.nan** values across data arrays.
- Continuous topographical parameters were transformed into ordinal multi-criteria evaluation (MCE) vulnerability scores scaled from 0 to 4.
- Localized drainage metrics were computed across administrative geometries into a weighted network density index mapping vector features per **KM squared**



4. Methodology

- Data Collection** — Five spatial datasets acquired: SRTM elevation raster (NASA EarthExplorer), WorldPop population raster, flood hazard index and incident records 2016–2020 (DKI Jakarta Open Data), and drainage network (OpenStreetMap via OSMnx)
- CRS Alignment** — All layers reprojected to a common coordinate reference system for spatial consistency
- Raster Processing** — Elevation tiles merged, rasters clipped to Jakarta mainland boundary, NoData values removed, zonal statistics extracted per kecamatan
- MCE Scoring** — Each indicator scored 0–4 against absolute thresholds and weighted: $0.45 \times \text{Incidents} + 0.30 \times \text{Hazard} + 0.15 \times \text{Elevation} + 0.10 \times \text{River Proximity}$
- Spatial Weights Matrix** — Queen contiguity matrix constructed ($n=42$, avg 5.1 neighbours), row-standardised
- Spatial Autocorrelation** — Global Moran's I, Local Moran's I (LISA), and Getis-Ord G_i^* applied to detect clusters and hotspots
- Spatial Regression** — OLS baseline with LM diagnostics, followed by ML Spatial Lag model to account for neighbourhood spillover
- Infrastructure Gap Analysis** — Drainage density computed per km^2 , overlaid with risk tiers to identify priority investment zones
- Shelter Placement** — Spatial neighbour algorithm identifies safe adjacent districts for emergency shelter siting

5. Results & Discussion



Acknowledgements: Please add acknowledgements here.